

Michael Marchev

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EDUCATION

Northeastern University, College of Engineering

Boston, MA

Candidate for BS in Mechanical Engineering and Design

May 2028

Awards & Activities: Generate - Hardware, Northeastern Electric Racing, Dean's List

GPA: 3.34

Coursework: FEA & Design, Dynamics, Mechanics of Materials, Thermodynamics, Statics, Materials Science

PROFESSIONAL EXPERIENCE

Stryker (Endoscopy)

San Jose, CA

R&D Mechanical Engineering Co-op

January 2026-June 2026

- Designed **<10 μ m** optical positioning bench in **4 motorized + 4 manual axes** with full blackout cage including CNC/manual milling, assembly, testing, supplier management, and control documentation
- Integrated HALT and standard/torsional Instrons with 3D printed/machined one-off fixtures
- Led Verification & Validation testing; vibration, thermal, thermopile, mechanical reliability
- Optimized prototype manufacturing methods to reduce materials **spending by 57%**

Richey & Clapper Inc.

Sudbury, MA

Power Equipment Mechanic - STIHL Certified Silver Technician

May 2025-August 2025

- Fully rebuilt 2 & 4 stroke petrol/diesel/electric motors, transmissions, gearboxes, carburetors, and serviced belts, hydraulics, and tires to support full range of contractor-grade lawn machinery
- Perform cost & failure mode analysis to determine repair procedure and custom parts modification

PROJECTS AND LEADERSHIP

Generate Product Development Studio

Boston, MA

Mechanical Technical Lead - Automated X-Ray Survey Machine

July 2026-Present

- Design top-level architecture of **~200 ft³ 5-axis** motorized gantry for detecting and logging errant X-ray emissions of industrial CT scanners, **saving ~1 hr** operator time per unit
- Run team meetings, assign action items, and direct subsystem design for **4 mechanical engineers**
- Manage integration between mechanical, electrical/firmware, and user interaction systems

Mechanical Engineer - Sageware

September 2025-December 2025

- Automated reliable fabric upcycling through compact process-based bead production solution
- Formulated robust cutting process with machined steel stamp and SLA resin fixture, linearly actuated by stepper motor, ball screw, and linear rail integrated with molding + dispensing systems

Mechanical Engineer - Uplift Solutions

December 2024-April 2025

- Iteratively developed stiff conduit linkages on powered walker with raising/tilting seat and legs

Northeastern University Center for STEM Education - Young Scholars' Program

Boston, MA

Program Co-Coordinator

June 2024-August 2024

- Managed 6-week NSF-funded university research program for 27 high school seniors
- Coordinated experiential education, tutored on professional development/communication skills

Research Assistant - ***Published in Annals of Biomedical Engineering (Vol. 53)***

June 2023-August 2023

- Researched impact mechanics and force propagation in advanced combat helmet systems
- Quantified precompression of helmet padding via experimental testing (**89 subjects**) for FEA

LSRHS Math Team

Sudbury, MA

Captain and Varsity Competitor

September 2020-June 2024

- Ran weekly meetings of **70+ members**, led team to **7th place** in MML, tutored, created/ran tryouts

VolunTeachable.com

Boston, MA

Co-Founder

Sep 2020 – Aug 2024

- Founded volunteer-based global linguistic exchange; taught **500+** underprivileged children/schools

CERTIFICATIONS + SKILLS

Certifications: CSWA, CSWA-AM, CSWA-Sustainability, STIHL Certified Silver Technician, Lean Green Belt

CAD/Manufacturing: SOLIDWORKS, Creo, Onshape, FDM + SLA + MJP 3D Printing, 3D Slicer, DFM, DFA

Skills: Product/Part Design, Engine Repair, Shop + Hand Tools, C++, Python, MATLAB, InDesign, Photoshop

Languages: English: *Native*, Bulgarian: *Fluent*, Mandarin: *Conversational*

Interests: Coffee, Judo, Audio/Music, Hiking, Skiing, Running, Violin, Piano, Soccer, Wellness, FDM